

Algorithms HW 2

1.19.12

Braden Buchanan

To calculate the power of a number faster than $O(n)$, the factors of the power must be determined. Take the power and keep dividing by each prime number (2, 3, 5, 7, etc.). Each division with no remainder means a factor has been found. Start over after each found factor.

```
List<int> factors = new List<int>();
int baseNum, int power, int total;
baseNum = 2; power = 5; total = 1;
// potentially  $O(n^2)$ ?
```

```
for (int i = 2; i < power/2; i++)
{
    if (power % i == 0)
    {
        factors.add(i);
        i = 2;
    }
}
```

```
// Loop through factors to determine total
foreach (int factor in factors)
{
    total *= factor;
}
```